

Climate Survival

Technical Architecture Document

Prepared for Prospective Investors

Version 1.0 | June 29, 2026

1. Executive Summary

Climate Survival is a climate-adaptive intelligence platform that helps individuals and communities make data-driven decisions about what food and supplies to stockpile before climate-driven price surges, which crops to plant that will survive forecasted weather anomalies, and when to buy based on real-time market price data from local sources.

The platform aggregates 8+ free public data sources including global climate models, satellite-derived hydrometeorological data, national market price reports, and product imagery. Data is processed through a hybrid rule-engine + Gemini AI pipeline and delivered via a polished web interface.

Built on zero-operating-cost infrastructure (Netlify + Hugging Face Spaces) and monetized through Daraz affiliate commerce in Sri Lanka, Climate Survival is designed for rapid global scalability with no cloud vendor lock-in.

2. System Architecture Overview

The system follows a decoupled architecture with a static frontend served via Netlify CDN and a backend API running in a Hugging Face Spaces Docker container.

Frontend Layer: Next.js 16 + React 19 static site on Netlify CDN (\$0/month)

Backend Layer: .NET 9 Web API Docker container on Hugging Face Spaces (\$0/month)

Data Sources: 8+ free public APIs (Open-Meteo, BigDataCloud, ip-api.com, World Bank, Wikipedia, Google Gemini, CBSL PDF)

Monetization: Daraz Affiliate Program (commission-based, Sri Lanka only)

3. Technology Stack

Frontend: Next.js 16.2, React 19.2, TypeScript 5.x, Tailwind CSS v4, Recharts 3.9, jsPDF + html2canvas

Backend: .NET 9 ASP.NET Core, Polly (resilience), MailKit (SMTP), PdfPig (PDF parsing)

AI: Google Gemini 2.0 Flash (free tier, 60 req/min) for LLM-powered crop recommendation re-ranking

4. External API Integrations (All Free)

Open-Meteo Seasonal: 3-month climate anomalies and Extreme Forecast Index

Open-Meteo ECMWF: 16-day weather forecast ensemble

Open-Meteo WMO/ERA5: Hydrometeorological data, soil moisture, flood risk

BigDataCloud: Reverse geocoding (1,000 free req/day)

ip-api.com: IP-based country detection (45 req/min)

World Bank API: Country names and codes

Wikipedia API: Product and crop thumbnail images

Google Gemini: AI crop recommendation re-ranking

CBSL Sri Lanka: Daily market price PDFs (public government data)

5. Backend Architecture

Services: ClimateService, HydroMeteoService, CropRecommendationService, GeminiService, RecommendationService, LocationService, CountryLocaleService, PriceService, SriLankaPriceService, EmailService

API Endpoints: GET /api/recommendations, GET /api/crops, GET /api/locations/*, GET /api/prices/sri-lanka, POST /api/contact

The recommendation engine uses a hybrid architecture: a deterministic rule engine detects climate signals, then Google Gemini re-ranks crop candidates with suitability scores (0-100), plant-by dates, and growing advice. Graceful degradation to rule-only when Gemini is unavailable.

6. AI / Machine Learning Integration

Signal Detection Rules:

Temperature anomaly > 1.5C and precipitation < -20mm = El Nino.

Temperature anomaly < -1.0C and precipitation > 30mm = La Nina.

Precipitation anomaly < -30mm = Drought.

EFI temperature > 0.7 = Heatwave.

Flood risk > 50 = Flood Risk.

Drought severity > 60 = Extreme Drought.

Model: gemini-2.0-flash (free tier) with structured JSON output via prompt engineering. Graceful degradation to rule-only when API unavailable.

7. Data Flow & Request Lifecycle

1. User visits site ->
2. IP country detection ->
3. Locale selection ->
4. Location selection ->
5. API request to /api/recommendations ->
6. Backend orchestrates 5 parallel data sources (geocoding, climate anomalies, hydrometeo data, rule engine, Gemini AI) ->
7. Unified JSON response ->
8. Rich UI rendering with animations ->
9. User can download PDF, switch language, view CBSL prices, click affiliate links, or contact via form

8. Deployment & Cost Analysis

All services are on free tiers. Total monthly cost: \$0.00.

Netlify Free: 100GB bandwidth, 300 build minutes, auto HTTPS, CDN.

Hugging Face Spaces Free: 2 vCPU, 16GB RAM, 50GB storage.

Git workflow: Three remotes - origin (GitHub primary),

netlify-repo (auto-deploys frontend),

hf (auto-deploys backend Docker container on HF Spaces). Push-to-deploy in under 5 minutes.

Monthly Cost Breakdown

Service	Cost	Notes
Netlify Free	\$0	100GB bandwidth, 300 build minutes
HF Spaces Free	\$0	2 vCPU, 16GB RAM, 50GB storage
Open-Meteo APIs	\$0	No rate limit
Google Gemini Free	\$0	1,500 requests/day
ip-api.com Free	\$0	45 requests/min
BigDataCloud Free	\$0	1,000 requests/day
Wikipedia API	\$0	No rate limit
CBSL PDF	\$0	Public government data
TOTAL	\$0/month	Fully operational at zero cost

9. Monetization Strategy

Primary: Daraz Affiliate Program (Sri Lanka). Member ID: 155412816. Contextual affiliate links embedded in product recommendations with FTC disclosure compliance.

Expansion Opportunities: Amazon Associates (US/UK/EU), Flipkart (India), Shopee (SE Asia), Premium subscription tier with extended forecasts and SMS alerts, B2B API for agricultural businesses and crop insurance.

10. Localization System

Languages: English (en), Spanish (es), French (fr), Chinese (zh), Sinhala (si). Auto-detection via IP address through ip-api.com and CountryLocaleService. Manual override via LanguageSwitcher component at any time.

11. Competitive Advantages

Zero operating cost: \$0/month vs. typical \$50-500/month for competitors.

AI-powered crop recommendations: Gemini LLM re-ranking vs. rule-based only.

Local market data: CBSL daily prices for Sri Lanka vs. no local pricing data.

Multi-language: 5 languages with auto-locale detection vs. English-only.

Non-intrusive monetization: Affiliate commerce vs. ads or subscriptions.

Data diversity: 8+ integrated free APIs vs. 1-2 paid sources.

Zero infrastructure cost: Push-to-deploy in under 5 minutes with no cloud bills.

12. Scalability Roadmap

Near-Term (3 months): Expand product image coverage, Amazon Associates for non-LK markets, performance optimization.

Short-Term (3-6 months): User accounts with saved locations, email/SMS alerts, PWA mobile app, historical climate comparison.

Medium-Term (6-12 months): Premium subscription tier, B2B API for agricultural businesses, custom ML models for item recommendations, multi-region weather station integration.

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